

CHAPTER – 3

OPERATING SYSTEM

BASIC CONCEPTS

An operating system (OS) is a crucial software component that manages hardware and software resources on a computer or mobile device. It provides an interface between users and the computer hardware, enabling users to run applications and utilize hardware resources efficiently.

Key functions of an operating system include:

1. **Resource Management:** Allocating and managing hardware resources such as CPU, memory, disk space, and peripherals among different applications and users.
2. **Process Management:** Handling and executing processes or tasks, including scheduling tasks, managing their execution, and facilitating communication between processes.
3. **Memory Management:** Allocating and managing system memory (RAM), ensuring efficient utilization and providing virtual memory capabilities when physical memory is insufficient.
4. **File System Management:** Organizing and managing files and directories on storage devices, handling file operations such as creation, deletion, and access control.
5. **Device Management:** Managing communication between software and hardware devices, including drivers for peripherals like printers, keyboards, and network adapters.
6. **User Interface:** Providing a user-friendly interface for interaction, which can be command-line interfaces (CLI) or graphical user interfaces (GUI), allowing users to interact with the system and applications.
7. **Security and Access Control:** Ensuring system and data security through authentication, authorization, encryption, and other security measures to protect against unauthorized access and malicious attacks.
8. **Networking:** Supporting networking capabilities by managing network connections, protocols (like TCP/IP), and communication between devices on a network.
9. **Error Handling:** Detecting and handling errors and exceptions that occur during system operation, ensuring system stability and reliability.
10. **System Services:** Providing various system services such as time management, printing, backup, and recovery to support user applications and system functionality.



Operating systems come in various types, including Windows, macOS, Linux, Unix, Android, iOS, and more specialized systems like real-time operating systems (RTOS) for embedded devices and servers. Each type of OS has specific features, user interfaces, and applications that cater to different computing needs and environments.

3.1 Main Functions of an Operating System

1. Resource Management:

Resource management is a critical function of an operating system, encompassing the allocation and management of the computer's hardware resources such as the CPU, memory, disk space, and peripherals.

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The operating system ensures that each application gets the necessary resources while maintaining overall system efficiency. This involves prioritizing tasks, balancing load across multiple processors, and managing memory usage to prevent conflicts and optimize performance. By efficiently allocating resources, the operating system maximizes hardware utilization, reduces idle times, and ensures that applications run smoothly and responsively.

Key Points:

- Allocates CPU, memory, disk space, and peripherals.
- Prioritizes tasks and balances load.
- Manages memory usage to prevent conflicts.
- Maximizes hardware utilization and system efficiency.

2. Process Management:

Process management involves the handling and execution of processes, which are individual tasks or programs running on a computer. The operating system is responsible for creating, scheduling, and terminating processes. It ensures that processes are executed efficiently by assigning CPU time, managing process states (running, waiting, blocked), and handling context switches between processes. Additionally, the OS facilitates inter-process communication (IPC) to enable processes to exchange data and synchronize their actions. Effective process management ensures that multiple applications can run concurrently without interfering with each other, thereby providing a smooth multitasking experience.

Key Points:

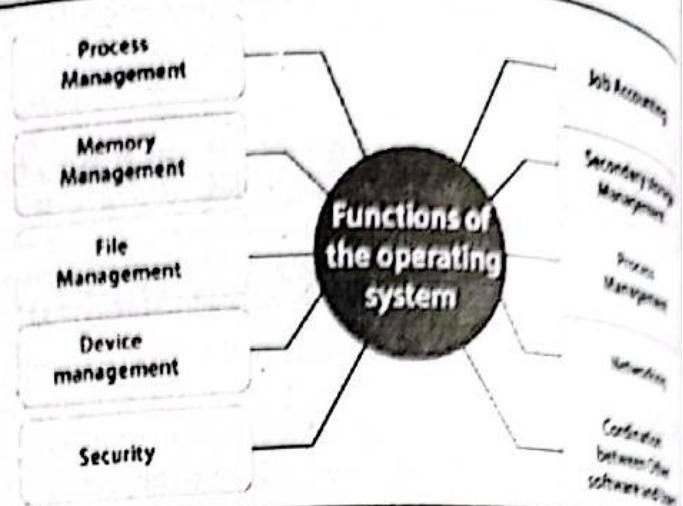
- Handles process creation, scheduling, and termination
- Manages process states and context switches
- Facilitates inter-process communication (IPC)
- Ensures efficient execution of concurrent applications

3. Memory Management:

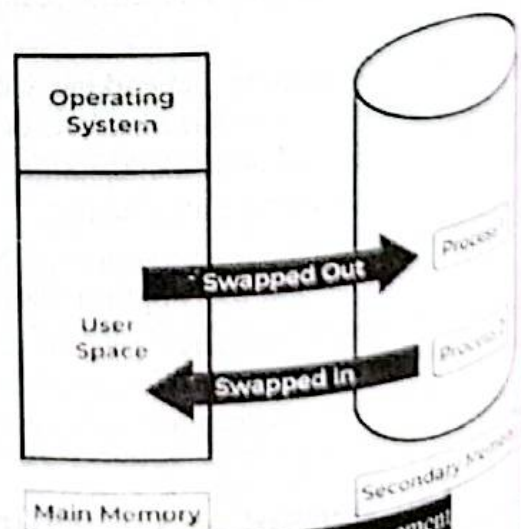
Memory management is the function that controls and coordinates computer memory, assigning blocks to various running programs to optimize overall system performance. The operating system handles both physical memory (RAM) and virtual memory. It tracks each byte in a computer's memory and allocates memory to processes while ensuring that the available memory is used effectively and efficiently. The OS employs techniques like paging and segmentation to manage memory, and it handles swapping between physical and virtual memory to accommodate larger programs. Effective memory management prevents memory leaks and fragmentation, ensuring the system remains stable and efficient.

Key Points:

- Manages physical and virtual memory



3.1 Functions of the operating system



3.2 Memory Management

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- Allocates memory to processes and tracks memory usage
- Uses paging and segmentation techniques
- Prevents memory leaks and fragmentation

4. File System Management:

The file system management function of an operating system organizes and controls how data is stored and retrieved on a storage device. It provides a hierarchical structure to store files and directories, making data access and organization intuitive for users and applications. The operating system manages file operations such as creation, deletion, reading, writing, and permissions. It also handles storage allocation, ensuring efficient use of disk space and preventing data loss. Advanced file system features like journaling help maintain data integrity by recording changes before they are committed to the main file system.

Key Points:

- Organizes and manages files and directories
- Handles file operations (creation, deletion, read/write, permissions)
- Manages storage allocation and disk space utilization
- Ensures data integrity with features like journaling

5. Device Management:

Device management is the function of the operating system that manages hardware devices and peripherals, such as keyboards, mice, printers, and network adapters. The OS provides a layer of abstraction between the hardware and the applications, allowing software to interact with hardware devices without needing to know the specifics of the device. It includes device drivers, which are specialized programs that facilitate communication between the OS and hardware devices. The OS also manages I/O operations, ensuring that data is transferred efficiently and correctly between devices and the system.

Key Points:

- Manages hardware devices and peripherals
- Provides abstraction layer for hardware interaction
- Includes device drivers for communication with hardware
- Manages I/O operations for efficient data transfer

6. User Interface:

The user interface (UI) function of an operating system provides the means through which users interact with the computer system. This can be a graphical user interface (GUI), which offers visual elements like windows, icons, and buttons, or a command-line interface (CLI), which relies on text-based commands. The UI facilitates user interaction with the system and applications, making it easier to perform tasks such as launching programs, managing files, and configuring system settings. A well-designed user interface improves user experience by being intuitive, responsive, and accessible.

Key Points:

- Facilitates user interaction with the computer system
- Can be graphical (GUI) or text-based (CLI)
- Helps perform tasks like launching programs and managing files
- Improves user experience through intuitive design

7. Security and Access Control:

Security and access control are crucial functions of an operating system, designed to protect the system and data from unauthorized access and threats. The OS implements authentication mechanisms to verify user identities and authorization to control access levels for different users and applications. It employs encryption to safeguard sensitive data and uses security protocols to protect against malware and

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cyberattacks. Additionally, the OS maintains access control lists (ACL) and permissions to ensure that users and applications only access resources they are permitted to, thereby maintaining system integrity and confidentiality.

Key Points:

- Protects system and data from unauthorized access
- Implements authentication and authorization mechanisms
- Uses encryption to safeguard data
- Maintains access control lists (ACL) and permissions

8. Networking:

The networking function of an operating system manages network connections and communications, enabling devices to connect and communicate over local and wide-area networks. The OS handles network protocols, such as TCP/IP, to ensure reliable data transmission between devices. It manages network interfaces, configures network settings, and facilitates data exchange through sockets and ports. Networking capabilities allow for resource sharing, internet access, and remote communication, making it possible for users to connect to and interact with other systems and services over a network.

Key Points:

- Manages network connections and communications
- Handles network protocols like TCP/IP
- Manages network interfaces and settings
- Facilitates data exchange and resource sharing over networks

9. Error Handling:

Error handling is a function of the operating system that detects, manages, and responds to errors that occur during system operation. The OS monitors system activities and resources, identifying issues such as hardware failures, software bugs, and resource conflicts. When an error is detected, the OS logs the error details, attempts to recover from the error, and may notify the user or system administrator. Effective error handling ensures that the system remains stable and reliable, minimizing the impact of errors on system performance and user experience.

Key Points:

- Detects, manages, and responds to system errors
- Monitors system activities and resources for issues
- Logs error details and attempts recovery
- Ensures system stability and reliability

10. System Services:

System services are essential background processes that the operating system provides to support the functionality of user applications and overall system operations. These services include time management, printing, system updates, backup and recovery, and various utilities that perform routine maintenance tasks. The OS ensures that these services run efficiently and are accessible to applications and users as needed. By managing system services, the OS enhances system functionality, improves performance, and ensures that critical tasks are performed reliably.

Key Points:

- Provides essential background processes
- Supports functionality of user applications
- Includes services like time management, printing, and system updates
- Enhances system functionality and performance

These detailed explanations and key points cover the main functions of an operating system, providing a comprehensive understanding of its role in managing computer resources and facilitating user interaction.

- **Resource Sharing:** Enables better utilization of system resources to run multiple applications simultaneously.
- **Scalability:** Facilitates running complex applications that require concurrent processing.

Challenges of Multitasking:

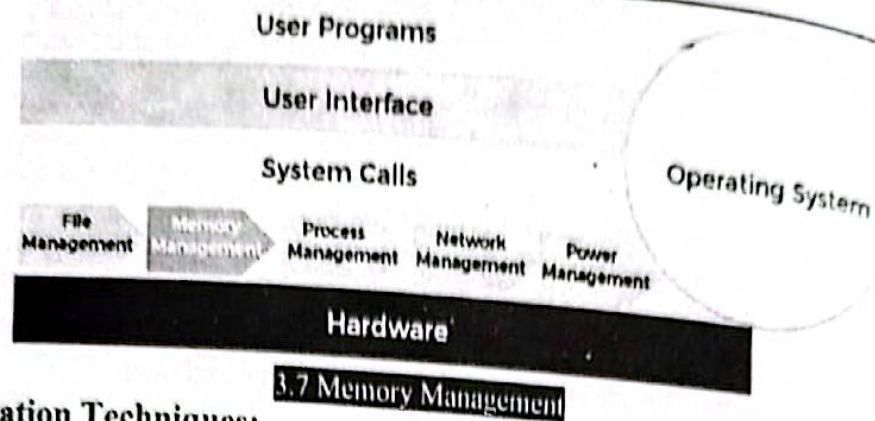
- **Resource Contention:** Multiple tasks competing for the same resources can lead to bottlenecks.
 - **Deadlocks:** Two or more tasks waiting indefinitely for resources held by each other.
 - **Synchronization Issues:** Ensuring data consistency and preventing race conditions.
 - **Overhead:** Context switching and scheduling can introduce performance overhead.
- Multitasking is a fundamental feature of modern operating systems, enabling efficient and effective use of computer resources while providing a seamless user experience.

MEMORY MANAGEMENT

Memory management is a critical function of an operating system (OS) that involves handling and optimizing the use of a computer's memory. The OS is responsible for allocating memory to processes, ensuring efficient utilization, and managing memory hierarchy (cache, main memory, and secondary storage). Here are the main aspects and techniques of memory management:

Key Aspects of Memory Management:

1. **Memory Allocation:**
 - **Static Allocation:** Memory is allocated at compile time and remains fixed throughout the program's execution.
 - **Dynamic Allocation:** Memory is allocated and deallocated at runtime based on the program's needs.
2. **Memory Protection:**
 - Ensures that processes do not interfere with each other's memory space.
 - Uses mechanisms like base and limit registers, segmentation, and paging.
3. **Memory Hierarchy:**
 - **Cache Memory:** Fast, small-sized memory close to the CPU.
 - **Main Memory (RAM):** Primary volatile storage for active processes.
 - **Secondary Storage:** Non-volatile storage like hard drives and SSDs for long-term data storage.
4. **Address Translation:**
 - Converts logical addresses generated by the CPU into physical addresses in memory.
 - Uses techniques like segmentation and paging.



3.7 Memory Management

5. Memory Allocation Techniques:

1. Contiguous Allocation:

- **Single Contiguous Allocation:** Entire memory space is allocated to a single process.
- **Partitioned Allocation:** Memory is divided into fixed or variable-sized partitions, each allocated to a different process.
- **Advantages:** Simple to implement.
- **Disadvantages:** Can lead to fragmentation.

2. Non-Contiguous Allocation:

- **Paging:** Divides memory into fixed-sized pages and divides processes into same-sized blocks called frames. The OS keeps a page table to map logical pages to physical frames.
 - **Advantages:** Eliminates external fragmentation.
 - **Disadvantages:** Can introduce internal fragmentation and overhead in maintaining page tables.
- **Segmentation:** Divides memory into variable-sized segments based on the logical divisions of a program (e.g., code, data, stack). Each segment is mapped to physical memory separately.
 - **Advantages:** Reflects logical structure of programs, can share segments.
 - **Disadvantages:** Can lead to external fragmentation.

Memory Management Techniques:

1. Virtual Memory:

- Allows the execution of processes that may not be completely in memory.
- Uses disk space to simulate additional RAM.
- Techniques like demand paging and swapping are used to bring pages in and out of memory as needed.

2. Demand Paging:

- Only brings pages into memory when they are needed, reducing memory usage and allowing more processes to run concurrently.
- **Page Fault:** Occurs when a required page is not in memory, causing the OS to load it from disk.
- **Page Replacement Algorithms:** Determine which pages to swap out when memory is full.
- **Examples:** FIFO (First-In, First-Out), LRU (Least Recently Used), Optimal Page Replacement.

3. Swapping:

- Entire processes are swapped in and out of memory to disk to free up memory for other processes.
- **Advantages:** Enables multiprogramming by sharing CPU time among multiple processes.
- **Disadvantages:** Can lead to significant overhead and reduced performance.

Fragmentation:

- 1. **External Fragmentation:** Occurs when free memory is scattered in small blocks throughout the system, making it difficult to allocate contiguous blocks of memory.

in and out of memory from executing actual tasks.

... more time swapping pages

Example of Virtual Memory Usage:

- A computer with 4 GB of RAM and a 20 GB hard drive might use virtual memory to create the illusion of a much larger pool of RAM. Programs that require more memory than is physically available can continue running by using the hard drive space as an extension of RAM.

Virtual memory is a crucial feature in modern operating systems, allowing for more flexible and efficient use of hardware resources while providing a stable and secure environment for running applications.

3.7 Comparison between Physical and Logical Addresses

Aspect	Physical Address	Logical Address
Definition	The actual address in the memory hardware (RAM).	The address generated by the CPU during program execution.
Visibility	Seen and used by the memory unit.	Seen and used by the CPU and applications.
Access	Directly accesses the memory cells.	Indirectly translated to physical addresses by the MMU.
Generation	Determined by the hardware (physical memory layout).	Generated by the CPU during runtime.
Mapping	Directly mapped to the hardware.	Mapped to physical addresses via the page table.
Flexibility	Fixed, as it depends on the actual hardware configuration.	Flexible, as it can be mapped to any physical address.
Security	Not inherently secure, can lead to memory conflicts.	Provides isolation between processes.
Usage	Used by the memory management unit (MMU) to access RAM.	Used by the CPU to execute instructions and access data.
Example	0 x 1A3F4C (a specific memory cell in RAM).	0x0045B6 (an address in the program's virtual address space).

This table should help clarify the differences between physical and logical addresses in a computer system.

3. Worst Fit:

- Allocates the largest available memory block that is large enough to accommodate a process.
- **Implementation:** The operating system searches for the largest block available to allocate to a process. This method aims to leave larger contiguous free spaces for future allocations.
- **Advantages:** Helps prevent premature memory fragmentation by leaving larger gaps between allocated blocks.
- **Disadvantages:** Slower allocation compared to first and best fit, can lead to inefficient use of memory due to larger leftover gaps.

These memory allocation techniques play a crucial role in determining how efficiently an operating system manages its memory resources, balancing performance, utilization, and fragmentation. Each technique has its strengths and weaknesses, making them suitable for different types of systems and applications.

FILE SYSTEMS

A file system is a crucial component of an operating system (OS) that manages the storage, retrieval, and organization of data on storage devices such as hard drives, SSDs, and removable media. It provides a systematic way to control how data is stored, accessed, and managed. Here's a detailed overview of file systems in operating systems:

Structure and Organization:

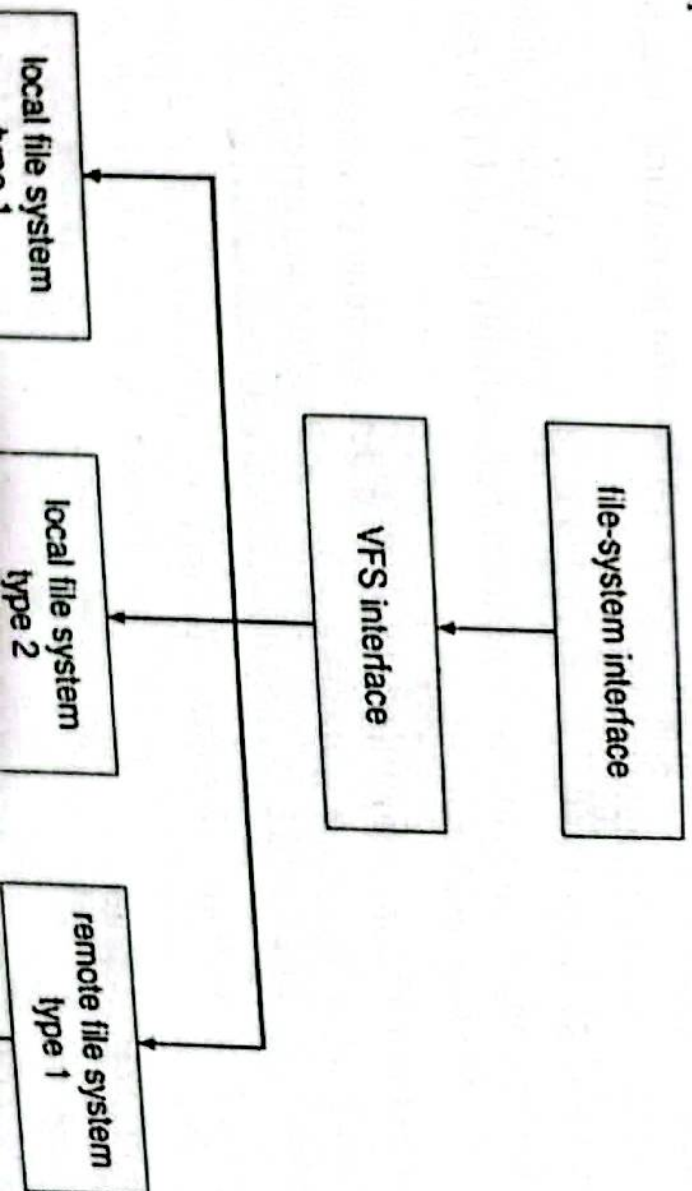
The file system is responsible for organizing files and directories in a hierarchical structure, typically starting with a root directory. This structure allows users to navigate through files and directories easily. Each file and directory is identified by a unique path, which specifies its location within the hierarchy. For instance, in Unix-like systems, the path `/home/user/documents/file.txt` indicates the `file.txt` located within the documents directory, which is in turn within the user directory under `home`.

File Allocation:

File allocation methods determine how the file system allocates space for files on the storage device. Common methods include:

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1. **Contiguous Allocation:** Allocates contiguous blocks of storage for a file. This method is simple and provides fast access times but can lead to fragmentation and difficulty in finding large enough contiguous spaces for new files.
2. **Linked Allocation:** Uses linked lists to allocate blocks of storage for files. Each block contains a pointer to the next block. This method eliminates fragmentation but can result in slower access times due to the need to traverse the pointers.
3. **Indexed Allocation:** Uses an index block to keep track of all the blocks allocated to a file. This method provides direct access to any block of a file, improving access times. However, it requires additional space for the index block.



the preemptive and non-preemptive scheduling has in the ability to interrupt a running process. Preemptive scheduling is more flexible and responsive, making it suitable for systems requiring quick reaction times and handling multiple interactive users. Non-preemptive scheduling, on the other hand, offers simplicity and predictability, making it ideal for environments where processes need to run to completion without interruption and where the overhead of context switching needs to be minimized.

In summary, the choice between preemptive and non-preemptive scheduling depends on the specific requirements of the operating environment, with preemptive scheduling favoring responsiveness and fairness, and non-preemptive scheduling favoring simplicity and low overhead.

INPUT/OUTPUT MANAGEMENT

I/O management is a critical component of an operating system, responsible for controlling and coordinating the input and output operations of the computer system. Efficient I/O management ensures that various I/O devices, such as keyboards, mice, printers, and storage devices, operate smoothly and interact effectively with the system's CPU and memory. Here are the key aspects of I/O management in an operating system:

I/O Hardware:

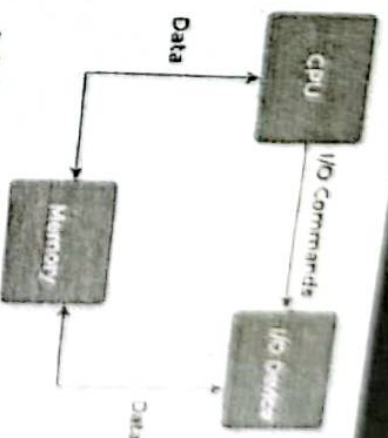
I/O devices are diverse, ranging from human-interface devices like keyboards and monitors to storage devices like hard drives and SSDs. Each device has its unique characteristics and communication protocols, making I/O management complex. The operating system must manage device drivers that serve as intermediaries between the hardware and the system's applications, translating high-level commands into device-specific operations.

Device Drivers:

Device drivers are specialized programs that allow the operating system to communicate with hardware devices. Each device type requires a specific driver to handle its operations. The driver abstracts the hardware details and provides a standardized interface for the operating system and applications. This modularity ensures that the system can support a wide range of devices without requiring changes to the core operating system code.

I/O System Calls:

I/O system calls are the mechanisms by which applications request I/O operations from the operating system. These calls provide functions for reading, writing, opening, closing, and manipulating files and devices. The operating system handles these calls, ensuring proper sequencing and coordination to avoid conflicts and ensure data integrity.



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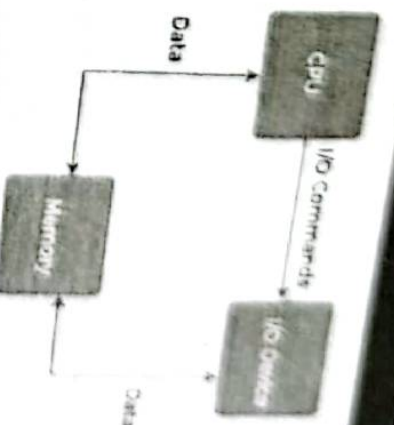
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Buffering:

Buffering is a technique used to manage data transfer between the CPU and I/O devices, which often operate at different speeds. Buffers are temporary storage areas that hold data while it is being transferred between devices and the CPU. This technique helps to smooth out variations in data transfer rates, improve system performance, and prevent data loss.

Caching:

Caching involves storing frequently accessed data in a fast storage medium to reduce access times. In I/O management, caches are used to store data temporarily, allowing quicker access to the most frequently used information. This technique improves overall system performance by reducing the need for repeated I/O operations.

Spooling:

Spooling (Simultaneous Peripheral Operations On-line) is a technique used to manage data that is sent to slow devices, such as printers. In spooling, data is first written to a buffer or spool file, from which the device can read at its own pace. This allows the CPU to continue with other tasks without waiting for the I/O operation to complete, improving system efficiency and resource utilization.

Interrupts:

Interrupts are signals sent by I/O devices to the CPU, indicating that they require attention. When an interrupt is received, the CPU pauses its current operations, saves its state, and executes an interrupt handler to address the device's needs. This mechanism allows for efficient handling of asynchronous I/O operations and ensures that devices receive timely responses.

Direct Memory Access (DMA):

Direct Memory Access (DMA) is a feature that allows I/O devices to transfer data directly to and from memory without involving the CPU for each byte of data transferred. This offloads the data transfer workload from the CPU, freeing it up to perform other tasks and improving overall system performance. DMA is particularly useful for high-speed data transfer devices, such as disk drives and network cards.

I/O Scheduling:

I/O scheduling involves determining the order in which I/O operations are executed to optimize performance and resource utilization. The operating system uses various algorithms to schedule I/O requests, balancing factors such as request urgency, device availability, and system throughput. Effective I/O scheduling minimizes wait times, reduces bottlenecks, and ensures fair resource allocation.

Error Handling:

I/O operations are prone to errors due to hardware failures, incorrect commands, or data corruption. The operating system includes mechanisms for detecting, reporting, and recovering from I/O errors. Error handling ensures data integrity and system stability, allowing the system to gracefully recover from faults and continue operation.

Conclusion:

Effective I/O management is essential for the smooth operation of an operating system, ensuring that various devices interact seamlessly with the CPU and memory. Through techniques like buffering, caching, spooling, interrupt handling, and DMA, the operating system optimizes I/O performance, improves system responsiveness, and ensures data integrity. Proper I/O management enhances the overall user experience and allows the system to support a wide range of applications and devices.

3.13 Device Drivers

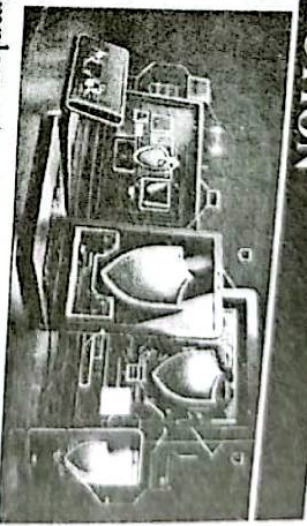
Device drivers are essential software components that facilitate communication between the operating system (OS) and hardware devices attached to a computer system. They serve as intermediaries, enabling applications and the OS to interact with hardware devices without needing to understand the specific details of each device's operation.

	input from the keyboard.	Network communication, GUI applications, non-blocking socket I/O.
Use Cases	Suitable for simple, sequential I/O tasks where waiting is acceptable.	Suitable for high-performance, real-time, and interactive applications requiring minimal latency.
Resource Management	Simpler resource management due to sequential operation handling.	More complex resource management to handle multiple concurrent operations.

This table highlights the primary differences between synchronous and asynchronous I/O, illustrating their respective advantages and trade-offs in various application scenarios.

SECURITY AND PROTECTION

Security in operating systems is a critical concern, as these systems serve as the foundation for all computing activities, managing hardware resources, executing applications, and facilitating communication between users and software. Here are detailed explanations of key aspects of security in operating systems:



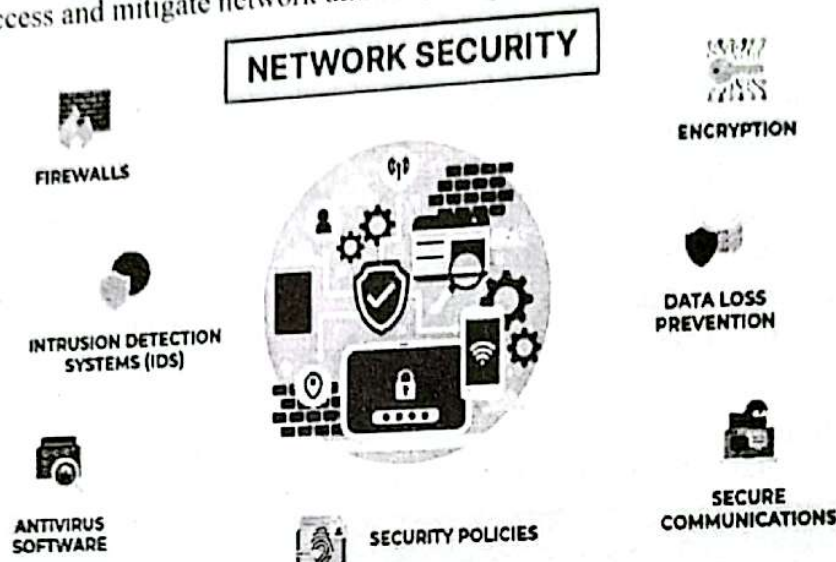
1. Access Control and Authorization: Operating systems implement access control mechanisms to ensure that only authorized users and processes can access resources. This involves assigning permissions to files, directories, devices, and network resources based on user roles and privileges. Access control lists (ACLs) and permissions such as read, write, execute, and ownership define who can access, modify, or execute specific resources. Effective access control helps prevent unauthorized access, data breaches, and malware attacks by enforcing least privilege principles—granting users only the permissions necessary for their tasks.

2. Authentication and Identity Management: Authentication verifies the identity of users and processes attempting to access the system or its resources. Operating systems use various authentication methods such as passwords, biometrics (fingerprint, facial recognition), smart cards, and two-factor authentication (2FA). Strong authentication mechanisms are crucial to prevent unauthorized access and identity theft. Identity management systems maintain and authenticate user identities across distributed systems, ensuring consistency and security in user access across networks and applications.

3. Encryption and Data Protection: Operating systems employ encryption to protect sensitive data stored on disks, transmitted over networks, or processed in memory. Encryption algorithms such as AES (Advanced Encryption Standard) and RSA (Rivest-Shamir-Adleman) ensure that data is scrambled into

unreadable formats without the correct decryption key. File encryption, disk encryption (e.g., BitLocker, FileVault), and secure communication protocols (e.g., TLS/SSL) safeguard against data breaches, eavesdropping, and tampering. Data protection mechanisms also include integrity checks and digital signatures to detect unauthorized modifications.

4. Secure Communication and Networking: Secure communication protocols and network security measures are essential for protecting data transmitted over networks. Operating systems support protocols like IPsec (Internet Protocol Security), SSH (Secure Shell), and VPN (Virtual Private Network) to establish encrypted connections and secure data exchange between systems. Firewall software controls network traffic, filtering incoming and outgoing data based on predefined security rules to prevent unauthorized access and mitigate network attacks (e.g., DDoS attacks, intrusion attempts).



3.20 Secure Communication and Networking

5. Auditing and Logging: Operating systems maintain audit logs to record system events, user activities, and security-related incidents for forensic analysis and compliance purposes. Auditing tools monitor access attempts, privilege changes, file modifications, and network activities, providing visibility into potential security breaches or policy violations. Security information and event management (SIEM) systems aggregate and analyze log data from multiple sources to detect anomalies, identify security threats, and facilitate incident response and forensic investigations.

6. Patch Management and Vulnerability Mitigation: Regular updates and patches are critical to addressing security vulnerabilities and weaknesses in operating system components, applications, and third-party software. Operating system vendors release security patches and updates to fix known vulnerabilities, improve system stability, and enhance defense mechanisms against emerging threats (e.g., malware, zero-day exploits). Automated patch management systems streamline the deployment of updates across distributed systems while minimizing downtime and reducing the window of exposure to security risks.

7. User Education and Awareness: Effective security in operating systems also depends on user education and awareness programs. Users play a crucial role in maintaining security by following best practices, such as creating strong passwords, avoiding phishing scams, practicing safe browsing habits, and reporting suspicious activities. Training programs educate users about security policies, data protection measures, and incident response procedures to promote a culture of cybersecurity awareness and accountability across organizations and communities.